

InventOR: A Human-Reviewed, Prompt-Configured Large Language Model Decision Support System for Material-Level Inventory Control

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Abstract

Material planners in manufacturing must set a reorder point, a safety stock and an order quantity for thousands of plant-material pairs, usually under ERP parameters that were last revised long ago. We define, build and evaluate *InventOR*, a decision support system (DSS) for this recurring, semi-structured operational decision. The planner remains the decision-maker. For each material the system supplies a candidate reorder-point/order-quantity (r, Q) policy generated by a large language model (LLM) that is configured only through a prompt, without training or fine-tuning, together with a written justification, an escalation flag and a counterfactual simulation. Deterministic software parses every proposal, gates it against minimum order quantities and storage limits, and replays it on held-out post-cutoff demand before the planner sees it. We derive six design requirements and evaluate the artifact on 365 plant-material pairs from three plants of an industrial manufacturer (346 in the common-feasible cohort), using three independent LLM runs, four reference arms that differ

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in the information they use, and an ablation that withholds the ERP planning parameters. Against ERP-derived parameters the LLM policies cost about 11% less in simulation at a service level 0.1–0.4 percentage points lower. An operations-research (OR) policy computed from the history alone, however, costs about 6% *less* than the LLM at the same service, so much of the apparent gain reflects weak ERP benchmarks rather than optimization by the LLM; the saving comes from holding about 17% less inventory at about 13% more stockout days, and when shortages are priced at one unit cost or more the ERP-derived parameters become the cheapest arm. Withholding the ERP parameters leaves the LLM about 9% cheaper than ERP-derived parameters and about 9% more expensive than the history-only OR policy. Storage limits that appear satisfied in a static check are exceeded on 53–66% of pairs along unconstrained simulated trajectories, so a hard-limit replay is our primary evaluation. At one plant the LLM is 31–36% more expensive than ERP-derived parameters, a result that is robust to our calendar repairs. The evidence supports InventOR as a human-reviewed DSS whose value lies in producing an auditable candidate policy for every material, not in out-optimizing a competent OR baseline. We contribute design principles for LLM-based DSS, explicit trust, escalation and rejection rules, and a research agenda.

Keywords: Decision Support Systems, Large Language Models, Generative Artificial Intelligence, Inventory Control, Human–AI Decision Making, Design Science, Operations Research

1. Introduction

Replenishment parameters in manufacturing are set per plant-material pair: a reorder point that triggers an order, a safety stock that absorbs de-

mand and lead-time variability, and an order quantity. In large ERP systems these values often remain at levels set years earlier, while demand patterns, supplier performance and storage constraints change. Revisiting them materially requires analysts who can fit demand and lead-time statistics, apply inventory theory, respect minimum order quantities (MOQs) and storage limits, and explain the result to planners. That scarce skill is exactly what a conversational large language model (LLM) with a code interpreter appears to offer, and a rapidly growing literature explores such tools for supply-chain and operations decisions [1, 2, 3, 4, 5].

The promise is real, but so are three problems that limit its use in practice. First, LLM outputs can be unfaithful or fabricated [6], so a proposed policy must be treated as a candidate and not a decision. Second, claims of superiority over ERP parameters are hard to interpret when the LLM and the comparators are given different information; an LLM that sees a plant’s ERP reference values and is compared with an OR policy that does not, or the reverse, tells us little about the LLM. Third, inventory policies live inside physical limits: a storage rule that holds when the policy is written down can be violated as stock builds up over time.

This paper treats the problem as one of decision support system (DSS) design. We define *InventOR* as a human-reviewed, model-driven DSS for the recurring decision of setting replenishment parameters per plant-material pair (Section 2). A planner remains the decision-maker. The system supplies, for each material, a candidate (r, Q) policy proposed by an enterprise LLM application that is configured only through a prompt and receives no project-specific training or fine-tuning, together with a justification, an escalation flag and a counterfactual simulation; deterministic software parses, gates and simulates every proposal before the planner reviews it and alone decides

31 whether any ERP parameter is changed. The pattern is not specific to
32 inventory: any production-planning decision in which parameters are set per
33 item and refreshed periodically (lot sizes, planned lead times, buffer levels,
34 dispatching rules) has the same structure of many semi-structured decisions,
35 scarce analyst time, hard constraints and a need for accountable evidence.
36 The design lessons below therefore concern LLM-based decision support for
37 production and logistics planning in general, and not only inventory control.

38 We follow a design-science logic [7, 8]: we state the problem and its scope,
39 derive design requirements, build the artifact, and evaluate it against those
40 requirements on real industrial data. Four research questions structure the
41 paper.

42 **RQ1. *Design.*** How should an LLM be embedded in a DSS so that its
43 proposals are bounded, verifiable, reproducible and reviewable by a
44 planner?

45 **RQ2. *Efficacy.*** What cost and service do the proposed policies achieve
46 relative to ERP-derived parameters and to OR policies, when the
47 information available to each approach is controlled?

48 **RQ3. *Feasibility and reliability.*** Do the policies respect physical storage
49 limits along simulated inventory trajectories, and how stable are they
50 across independent runs?

51 **RQ4. *Use.*** Under which conditions should a planner trust, escalate or reject
52 a proposal?

53 Our contributions are:

54 **C1. A definition and scope of the DSS** (Section 2): the decision,
55 the decision-maker, the system boundary, the level of automation and

56 six design requirements that make the role of the LLM explicit and
57 bounded.

58 **C2. A DSS architecture and operating rules** (Section 4) that separate
59 generation, deterministic gating, simulation and planner review, with
60 explicit rules for when output is trusted, escalated or rejected, and
61 design principles that generalize beyond this application (Section 8).

62 **C3. An information-controlled, trajectory-level evaluation** (Sec-
63 tion 5) with four reference arms that differ in the information they use
64 and, partly, in method, an LLM ablation denied the ERP planning
65 parameters, and a hard-limit storage replay that is the primary result.
66 This separates what the LLM adds from what it is given.

67 **C4. Complete-cohort industrial evidence** from three LLM runs on 365
68 plant-material pairs, with plant-level, regime-level and sensitivity anal-
69 yses, and managerial and research implications.

70 Our main findings are as follows. Against ERP-derived parameters, the
71 LLM policies are about 11% cheaper at a service level that is 0.1–0.4 points
72 lower. That advantage largely disappears when a reasonable OR baseline is
73 available: a history-only (r, Q) policy is about 6% cheaper than the LLM at
74 essentially equal service, and the LLM’s saving over ERP parameters comes
75 from carrying about 17% less inventory at about 13% more stockout days;
76 once shortages are priced at one unit cost or more, the ERP-derived pa-
77 rameters are the cheapest arm. The LLM’s value is therefore not superior
78 optimization but the ability to produce a reviewable, auditable policy for
79 every material without writing code. Storage limits are violated along un-
80 constrained simulated trajectories for 53–66% of pairs under every method,

81 which makes trajectory checks a prerequisite for any replenishment decision
82 aid. At one plant the LLM is 31–36% more expensive than ERP-derived
83 parameters, which we use to derive guidance on where to trust and where to
84 review.

85 The remainder of the paper is organized as follows. Section 2 defines
86 the problem, the scope of the DSS and its design requirements. Section 3
87 positions the work in the literature. Section 4 describes the architecture and
88 operation of the DSS for a non-specialist reader. Section 5 describes the
89 data and the evaluation design. Section 6 reports results. Section 7 derives
90 managerial insights, and Section 8 discusses design principles, limitations
91 and research directions. Formal definitions are in Appendix Appendix A.

92 **2. Problem, Scope and Definition**

93 *2.1. The decision problem*

94 The decision supported is the periodic (re)setting of three replenishment
95 parameters, a reorder point, a safety stock and an order quantity, for each
96 plant-material pair held in an ERP system. In the classification of Gorry
97 and Scott Morton [9] it is an operational-control decision that is partly struc-
98 tured: the calculation is textbook, but the choices of estimator, the treat-
99 ment of intermittent or trending demand, the plausibility of recorded lead
100 times and the handling of constraints require judgement that is applied in-
101 consistently when thousands of materials compete for a few analysts' time.
102 The decision-maker is the material planner or inventory analyst; the deci-
103 sion is repeated, high-volume and consequential for working capital, service
104 and storage. Hard constraints are the minimum order quantity (MOQ) and
105 a physical storage limit. The feedback loop is slow: a changed parameter

106 shows its effect only in subsequent demand.

107 2.2. Definition and level of automation

108 **Definition.** *InventOR is a human-reviewed, model-driven deci-*
109 *sion support system that, for one plant-material pair at a time, (i)*
110 *obtains from a large language model, configured only by a written*
111 *prompt, a candidate (r, Q) policy with a justification and an es-*
112 *calation flag; (ii) verifies the candidate deterministically against*
113 *explicit constraints; (iii) evaluates it by counterfactual simulation*
114 *against reference policies on held-out demand; and (iv) presents*
115 *the result to a planner, who alone decides whether ERP parame-*
116 *ters are changed.*

117 In the classical DSS architecture of data, model and dialog components
118 [10, 11], the LLM and the simulator form the model component, the material-
119 scoped data extract and the prompt form the data and knowledge compo-
120 nents, and the review interface is the dialog component (Section 4.1). In
121 terms of the levels of automation of Parasuraman et al. [12], InventOR op-
122 erates at a low-to-intermediate level: it selects and suggests one alternative
123 and leaves execution to the human. This is a design decision, not a limita-
124 tion: acceptance of automated suggestions is accompanied by complacency
125 and automation bias [13], which the review rules of Section 4.3 are meant to
126 counter.

127 2.3. Scope

128 Table 1 states what is and is not part of the system and of the evaluation,
129 so that the claims of the paper are read at the right level.

Table 1: Scope of InventOR and of its evaluation.

Dimension	In scope	Out of scope
Decision	Reorder point, safety stock and order quantity per plant-material pair, reviewed periodically	Order release, supplier selection, pricing, forecasting-model development
Inventory system	Single item, single echelon, continuous review, lost sales, (r, Q) family	Multi-item coupling, shared capacity, multi-echelon networks
Inputs	Daily demand and realized lead times, working-day flags, cost and constraint parameters, ERP reference parameters	External signals, supplier forecasts, market data
Role of the LLM	Candidate generator with justification and escalation flag	Forecasting, order placement, running the simulation, final decision
Human role	Planner reviews and decides; rules for trust, escalation and rejection	Autonomous execution without review
Evaluation	Ex-post simulation on real post-cut-off demand, information-controlled comparison, hard storage-limit replay, repeated runs	Field trial, planner workload or decision quality, user acceptance

130 2.4. Design requirements

131 Four sources motivate the requirements: the known unreliability of LLM
 132 outputs [6], guidance on human–AI interaction and accountability [14, 15],
 133 the run-to-run nondeterminism of LLM inference [16], and the practical con-
 134 straints of inventory control. They yield six requirements (Table 2), each
 135 tied to a design feature (Section 4) and an evaluation (Section 6).

Table 2: Design requirements, the design features that address them and where they are evaluated.

ID	Requirement	Design feature	Evaluation
DR1	Bounded generation: the LLM proposes and has no authority	Narrow candidate-generator role; prohibited methods; no order placement	Sections 6.1, 6.3
DR2	Verifiable feasibility along time, not only at a point	Deterministic gate; hard-limit trajectory replay	Sections 6.1, 6.5
DR3	Counterfactual evidence for the reviewer	Held-out simulation against reference arms	Sections 6.2, 6.8
DR4	Information transparency: comparisons control what each arm sees	Four reference arms; LLM ablation without ERP parameters	Sections 6.3, 6.4
DR5	Reproducibility and accountability of the configuration	Frozen prompt file; per-call log; repeated runs	Section 6.7
DR6	Human control with explicit escalation	Trust, escalate and reject rules; escalation flag	Sections 4.3, 6.6

136 2.5. Research approach

137 We follow the design-science research process of problem identification,
138 requirements, artifact design and evaluation [7, 8]. In the evaluation frame-
139 work of Venable et al. [17], our evaluation is artificial and ex post: it uses
140 real industrial data and a faithful simulation but no live users. It therefore
141 establishes technical efficacy, feasibility and reliability; it does not establish
142 usability, adoption or improved planner decisions, which we list as limitations
143 and research directions.

144 3. Related Work and Positioning

145 *Decision support systems and human-AI decision making..* DSS research
146 has long distinguished data-, model- and knowledge-driven systems and has
147 stressed that the value of a DSS lies in improving the judgement of a human
148 decision-maker rather than in replacing it [9, 10, 11]. LLMs add a new kind
149 of model component: it is configured through language, produces free-text
150 justification and is not deterministic. Research on human-automation in-
151 teraction shows that the way suggestions are scoped, presented and verified
152 determines whether users rely on them appropriately or over-rely on them
153 [12, 13], and guidelines for human-AI interaction call for explicit scoping,
154 explanation and correction mechanisms [14]. Our contribution to this stream
155 is a concrete, evaluated instance of an LLM-based DSS for an operational
156 production decision, with design requirements that are tied to evidence.

157 *LLMs in operations and supply-chain decisions..* Reviews and frameworks
158 map the opportunities for generative AI in supply chains [1, 2, 3, 4, 18]
159 and stress that adoption depends on assigned human roles, data quality and
160 governance [5, 19, 20, 21]. A recent conceptual line proposes agentic digital

161 twins that combine AI agents with model-based simulation and optimization
 162 to support decisions [22], and an LLM-plus-digital-twin method has been
 163 built for closed-loop supply-chain operations [23]. Our decision aid follows
 164 the same logic at a smaller scale: an LLM proposes and a simulation model
 165 checks. Most technical LLM work focuses on translating natural-language
 166 problem statements into optimization models and solver code [24, 25, 26, 27],
 167 on multi-agent control and consensus-seeking in inventory flows [28, 29], on
 168 extracting supply-network structure from text [30, 31], on disruption mod-
 169 elling [32], or on evaluating LLMs on classical operations-research tasks such
 170 as vehicle routing [33]. The closest LLM study to ours uses an LLM as a
 171 forecasting copilot for spare-parts demand, combined with simulation, and
 172 reports high variability for irregular demand that can be reduced by averag-
 173 ing repeated queries and enriching prompts [34]. We share the concern about
 174 run-to-run variability and test it directly (Section 6.7), but differ in that the
 175 LLM proposes a complete replenishment policy rather than a forecast.

176 *Data-driven inventory control without LLMs.* Learning-based replenish-
 177 ment is well established. Deep reinforcement learning has been applied to
 178 omnichannel and multi-echelon systems [35, 36] and aerospace replenishment
 179 [37]; deep-learning forecasting has been integrated with replenishment [38];
 180 contextual distributionally robust optimization uses side information to
 181 set order quantities [39]; and distribution-free online algorithms handle
 182 capacitated warehouses under uncertain price and demand [40]. Robust
 183 inventory control under demand and lead-time uncertainty has a longer
 184 history [41, 42, 43], and lead-time variability and dependence are known
 185 to propagate through ordering decisions [44], while safety stock and safety
 186 lead time are alternative buffers against unreliability [45]. The closest

operational neighbour is a trained data-driven framework that jointly sets replenishment parameters, safety stock and safety lead time on real automotive data [46]. These methods require model training, historical scenario generation or tuned hyperparameters for each setting; InventOR requires none, at the price of an LLM-based generator whose internal logic cannot be inspected beyond its written justification.

Intermittent and spare-parts demand. Material histories in manufacturing are frequently intermittent or lumpy. The statistical literature offers specialized estimators and classification schemes [47, 48, 49, 50, 51], and recent studies develop feature-based forecast combinations and temporal aggregation for intermittent demand [52, 53], forecasting for spare parts across product life-cycle phases [54], and integrated forecasting and planning or multi-echelon stocking models for repairable and spare parts [55, 56]. Our regime analysis (Section 6.6) shows where an LLM that is forbidden from using specialized intermittent-demand estimators behaves differently on lumpy materials.

Table 3 summarizes these streams and states how the contribution of InventOR differs. InventOR is not a new inventory model and does not claim to beat trained or optimized methods. It contributes (i) a DSS design in which an off-the-shelf LLM is a bounded candidate generator inside a reviewed process, (ii) an evaluation that controls the information available to each arm, and (iii) industrial evidence on cost, service and storage feasibility. The (r, Q) policy it uses is the standard workhorse of continuous-review control [57, 58, 59].

Table 3: Literature positioning of InventOR relative to recent work on AI-supported inventory and supply-chain decisions.

Study	Approach	Decision	Relation to InventOR
<i>LLMs for operations and supply chains</i>			
Jackson et al. [1]; Fosso Wamba et al. [2]	Framework / review	Broad supply-chain use	Motivates LLM use; no inventory-policy evidence
Song et al. [3]; Panja et al. [4]	Systematic reviews	LLMs in SCM and logistics	Place InventOR in the emerging empirical gap
Trifone et al. [5]; Cannas et al. [19]; Venkatesh et al. [20]	Adoption studies, search agenda	AI roles, barriers, data quality	Support human-in-the-loop design and review rules
Ivanov [22]; Le et al. [23]	Agentic LLM digital twins	Model-based decision support	Pair AI proposals with simulation; InventOR applies this to (r, Q) parameters
Nakhost et al. [34]	LLM + simulation	Spare-parts forecast	Closest LLM study; forecast, not policy
Jannelli et al. [29]; Quan and Liu [28]	Multi-agent LLM	Sequential ordering, consensus	Per-period agent decisions in simulation
AlMahri et al. [30]; Zheng and Brintrup [31]	LLM + knowledge graphs	Supply-network visibility	LLM for information extraction, not inventory control
Hamdan and Brahimi [33]	LLM benchmark	Routing	Evaluates LLM on a classical OR task
Li et al. [24]; Huang et al. [25]; Ahmadireshnizi et al. [26]; Zhang et al. [27]	NL→model solver	General OR models	Model formulation; not a fixed policy family
<i>Data-driven inventory control without LLMs</i>			
Gabellini et al. [46]	Trained data-driven	Replenishment, safety stock and safety lead time	Closest OR neighbour; needs training; automotive data
Kolyaei et al. [35]; Stranieri et al. [36]; Wang et al. [37]	Deep reinforcement learning	Replenishment	Learned policies; require training environments
Ma [38]; Wang et al. [39]; Zhang et al. [40]	Deep learning / contextual DRO / online algorithm	Forecast-plus-order; capacitated replenishment	Training, tuning or worst-case design per setting
<i>Intermittent and spare-parts demand</i>			
Sanguri et al. [52]; Li et al. [53]; Goldsmith and Sachs [54]	Statistical ML forecasting	Intermittent demand	Specialized estimators the LLM is not allowed to use
Babaveisi et al. [55]; Caserta and D’Angelo [56]	Optimization	Spare-part stocking	Multi-echelon or repairable settings; outside our single-echelon scope
Prak et al. [50]; Babai et al. [51]	Estimation review	Inventory under intermittency	Regime-aware benchmarks for future work
InventOR (this paper)	(this Prompted LLM + deterministic gate and simulation)	(r, Q) per plant-material pair	No training; reviewed; information-controlled evaluation

210 4. The InventOR Decision Support System

211 4.1. Architecture and components

212 Table 4 maps the classical DSS components [10, 11] to their realization in
 213 InventOR and to the design requirements of Section 2.4. The generator is the
 214 only component whose internal logic is not inspectable, and it is surrounded
 215 by components that are deterministic and auditable. In the evaluated proto-
 216 type the dialog component is the per-pair output record and the comparison
 217 tables; an interactive planner interface was not built or evaluated.

Table 4: DSS components of InventOR.

Component	Realization in InventOR	Requirement
Data	Material-scoped extract of daily demand, lead times, working-day flags, cost and constraint parameters, and ERP reference values (Table 5); the set of fields supplied is controlled and logged	DR4
Knowledge	System prompt file defining the task, the prohibited methods, the storage rules and the output schema (Supplementary File S1)	DR1, DR5
Model: generator	Enterprise LLM application with a code interpreter, configured only by the prompt; proposes one (r, Q) policy, a justification and an escalation flag	DR1
Model: verifier and simulator	Deterministic parser and gate; held-out simulation with hard storage-limit replay; reference arms	DR2, DR3
Dialog (review)	Policy, justification, flag and simulated comparison presented to the planner under the trust, escalate and reject rules	DR6
Governance	Frozen prompt file, per-call log of input hash, time, session and raw response, repeated runs	DR5

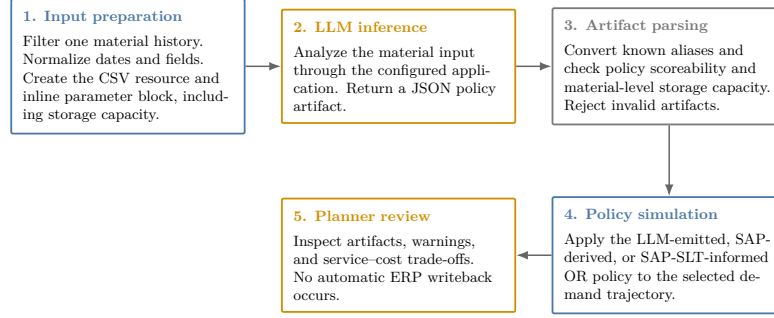
218 4.2. Operation in four steps

219 A planner’s question is: *given what this material has done in the past,*
 220 *how much safety stock should I hold, at what stock level should I reorder,*
 221 *and how much should I order?* InventOR answers this question in four steps
 222 (Figure 1).

223 **Step 1. Generate.** The LLM, accessed through an enterprise application
 224 with a code interpreter, receives one material’s history (daily de-

InventOR prototype workflow

Executed material-level path from prepared history to planner-reviewable simulation output.



Deterministic components reproduce simulation results from stored artifacts; LLM inference itself may vary.

Figure 1: InventOR workflow: material data are prepared, an LLM proposes a policy, deterministic software parses, checks and simulates it, and a planner reviews the result.

mand, realized lead times, working-day flags) and a short parameter block (unit cost, holding rate, ordering cost, MOQ, service target, storage limit, and current ERP safety stock and safety lead time for comparison). A written prompt (the only configuration; no weights are changed) [60] instructs it to compute demand and lead-time statistics, classify the demand pattern, apply an economic-order-quantity (EOQ) argument and a normal-approximation safety stock, respect the storage limit, and report a structured policy together with a short justification and escalation flag.

Step 2. Gate. Deterministic code extracts the policy, rejects malformed or infeasible outputs (missing or non-numeric fields, non-positive reorder point or order quantity, negative safety stock, reorder point below safety stock, safety stock plus order quantity above the storage limit) and retries the generation up to three times. The backtest applies one further admission check, that the order quantity is not

241 below the MOQ.

242 **Step 3. Simulate.** Each accepted policy is replayed on the material’s de-
243 mand observed *after* the data cutoff, a held-out evaluation in the
244 sense of out-of-sample forecast testing [61, 62], together with the
245 comparator policies, producing holding and ordering cost, fill rate
246 and stockout days. This plays the role that model-based simulation
247 plays in digital-twin decision support [22], at the scale of a single
248 policy.

249 **Step 4. Review.** A planner sees the policy, the LLM’s justification and
250 flag, and the simulated comparison. Only the planner changes ERP
251 parameters.

252 The role of the LLM is deliberately narrow. It is a *candidate generator*
253 whose output is never trusted without the gate, and it is not used to fore-
254 cast, to place orders, or to run the simulation. The prompt prohibits Monte
255 Carlo simulation, distribution fitting, machine-learning libraries and special-
256 ized intermittent-demand estimators, so that the policy rests on transparent
257 statistics a planner can reproduce.

258 4.3. Trust, review and reject rules

259 We propose three explicit rules for planners, in line with guidelines for
260 human–AI interaction and with internal-auditing practice for AI systems
261 [14, 15], and motivated by the evidence in Section 6.

262 **Trust (apply after review)** when the proposal passes the gate, the sim-
263 ulated trajectory respects the storage limit when orders are truncated at it,
264 and the simulated cost is not above that of a history-only OR reference with
265 a fill rate not more than about 0.5 points lower.

266 **Escalate** when the LLM raises an escalation flag, the material is lumpy or
267 has strongly seasonal or trending demand, the plant or calendar data were
268 repaired, or the policy differs strongly between repeated generations.

269 **Reject** when the proposal fails the gate after the retry budget, exceeds the
270 storage limit in the hard-limit replay, or is dominated (higher cost and lower
271 fill) by the history-only reference.

272 *4.4. Provenance and reproducibility of the configuration*

273 The system prompt, which fully defines the task, the forbidden methods,
274 the capacity rules and the output schema, is a plain-text file that is deployed
275 unchanged in the enterprise application; it is distributed with the paper
276 (Supplementary File S1) and the code repository. Every call logs the SHA-
277 256 of its input data, the UTC time, the application session identifier and
278 the full raw response. The platform does not expose the underlying model
279 identifier or sampling settings to the client; the evaluated model (Anthropic
280 Claude Sonnet 4.6, with a code interpreter) is therefore declared by the
281 operator and is not machine-verifiable. We report this limitation openly and
282 base the reproducibility argument on the prompt file, the per-call logs and
283 three independent runs (Section 6.7) rather than on a model checkpoint.

284 **5. Application and Evaluation Design**

285 *5.1. Evaluation approach*

286 The evaluation addresses each design requirement in turn (Table 2):
287 whether the pipeline returns valid, feasible policies (DR1, DR2); what cost
288 and service the policies achieve against reference arms on held-out demand
289 (DR3); how much of that result is due to the information given to the LLM

rather than to its reasoning (DR4); whether storage limits hold along the simulated trajectory (DR2); how stable policies are across independent runs (DR5); and where proposals should be escalated (DR6). The primary basis is the hard-limit replay with working-day receipts and no shortage penalty; sensitivity analyses vary these choices. The evaluation is ex post and uses no live users (Section 2.5).

5.2. Industrial application and data

The application is a pilot at a large manufacturer. The dataset contains 233,078 daily records for 411 plant-material pairs from three plants (A, B and C), with a data cutoff on 10 March 2026 and a post-cutoff horizon observed to 27 August 2026. Of the 411 pairs, 365 are eligible for scoring (234 in Plant A, 65 in Plant B, 66 in Plant C) and are sent to the LLM. For 19 pairs (all in Plant A) the supplied MOQ plus the reference safety stock exceeds the storage limit, so no comparator can satisfy both rules; these are excluded from outcome comparison, leaving a *common-feasible cohort* of 346 pairs (215, 65 and 66). All 365 pairs receive an LLM policy. Demand is concentrated: the five largest materials account for 40.3% of demand and the ten largest for 57.8%. The source working-day flag of Plant C is all zero; we reconstruct it from the same-date majority of Plants A and B (ties count as working days) and test the effect of this repair in Section 6.8. Such data-quality problems are a known threat to analytics-based decision support [63] and, for inventory systems, to record accuracy [64, 65, 66].

5.3. Reference arms and information content

The LLM and the classical comparators do not necessarily use the same information, and this matters for the interpretation of any cost difference,

Table 5: Fields of the material-scoped CSV given to the LLM.

Column	Type	Description and use
date	date	Observation date; time ordering and aggregation
actual_demand	numeric	Realized daily demand; mean, variability, intermittency, regime
delivery_time	numeric	Recorded replenishment lead time (receipt rows)
is_working_day	binary	Working-day flag; demand statistics over working days
minimum_order_quantity	numeric	MOQ; order-size constraint
unit_cost, holding_cost_rate, ordering_cost	numeric	Unit value, annual carrying-cost rate, fixed ordering cost; EOQ and cost evaluation
service_level_target	numeric	Target service used in safety-stock computation
max_storage_units	numeric	Storage limit: $MOQ \leq Q$ and $SS + Q \leq \text{max_storage_units}$
safety_stock_units, safety_time_workdays_actual	numeric	ERP planning parameters (current safety stock, safety lead time); withheld in the ablation arm
reliability_calc, supplier_plant_distance, transportation_network	mixed	Supplier and lane context

315 because ERP safety stock and safety lead time embed accumulated planner
316 judgement, and safety stock and safety lead time are alternative ways of
317 buffering unreliability [45]. Planned lead times can also drift upward when
318 they are adjusted dynamically (the “lead time syndrome” [67]), which would
319 inflate any comparator that adds them to the lead time. We therefore con-
320 struct arms that make the information explicit (Table 6).

Table 6: Comparison arms and the information each uses. All arms respect MOQ and are projected to the storage limit when possible.

Arm	Parameters from	ERP used	inputs	Purpose
SAP-derived	ERP safety stock; EOQ order quantity	Safety stock		Status quo of the plant
SLT-informed OR	Stationary (r, Q) with ERP safety lead time added to the lead time	Safety lead time		Original OR comparator
ERP-floor OR	SLT-informed OR with safety stock floored at the ERP value	Both		OR arm matched to the LLM’s full information
History-only OR	Stationary (r, Q) from demand and lead-time history, mean lead time	None		ERP-independent baseline
LLM (runs 1–3)	LLM proposal with all fields of Table 5	Both		Main treatment
LLM without ERP	LLM with ERP safety stock and safety lead time set to “not supplied”	None		Ablation: what the LLM does when it cannot anchor on ERP

321 5.4. Simulation, capacity replay and metrics

322 Each pair is simulated independently on the post-cutoff horizon as a
 323 single-item, single-echelon continuous-review (r, Q) system [57, 68, 69, 59, 58]
 324 with lost sales [70], in the spirit of simulation-based experiments on supply
 325 and timing uncertainty [71], a shared opening stock, an empty pipeline and
 326 arrivals after the rounded mean lead time in working days (Appendix Ap-
 327 pendix A). MOQs are treated as hard order-size constraints, as imposed
 328 by suppliers [72], and storage limits as capacity bounds on the replenish-
 329 ment decision [40]. The primary metrics are the mean material-horizon
 330 holding-plus-ordering cost, the mean material-level fill rate, the aggregate
 331 (volume-weighted) fill rate and the number of stockout days. We report
 332 paired differences with 10,000-sample bootstrap 95% intervals.

333 A static check $SS + Q \leq \text{max_storage_units}$ does not show that storage
 334 limits are respected as the inventory evolves, because the reorder point in-
 335 cludes lead-time demand and inventory on order can exceed the limit when
 336 it arrives. We therefore record, for every policy and every simulated day, the
 337 post-receipt on-hand stock and report the share of pairs and days above the
 338 limit (the *unconstrained* simulation). We then *replay* every policy with each
 339 order truncated to the storage limit minus the current inventory position
 340 (the *hard-limit replay*). The hard-limit replay is our primary evaluation.

341 5.5. Repeated runs and sensitivities

342 To examine run-to-run variation, the entire cohort is generated three
 343 times (runs 1, 2 and 3) with the same prompt file and the same data. Run 1
 344 serves as the reference for tabulated results unless stated otherwise; runs 2
 345 and 3 provide replicates. Sensitivity analyses vary the working-day recon-
 346 struction (majority, any-plant, strict-all, exclusion of Plant C and calendar-

day arrivals), the shortage penalty (zero, one and five times unit cost), and the exclusion of diagnostic outliers (52 pairs with coefficient-of-variation error above 0.75 or safety stock outside half to twice the reference).

6. Results

6.1. Completion and policy characteristics

All four runs (three with the full input and one ablation, Section 6.4) return a valid, storage-feasible (r, Q) policy for all 365 pairs after at most three attempts per pair. Retries were mostly caused by schema errors (invalid policy), not by capacity violations. The validity gate rejects non-positive reorder points and order quantities, negative safety stocks and reorder points below the safety stock. The ERP safety stock is zero for 198 of the 365 pairs (188 of the 346 in the cohort below), so for a majority of pairs the SAP-derived arm holds no safety stock at all.

6.2. Headline comparison with ERP-derived parameters

Unless stated otherwise, all results from here on use the hard-limit replay (every order truncated at the storage limit) on the 346-pair common-feasible cohort, with working-day receipts and no shortage penalty; Section 6.5 reports the unconstrained simulation for comparison. Table 7 compares all arms. In run 1 the LLM policies cost 11.4% less than SAP-derived policies and 30.5% less than the SLT-informed OR comparator (mean 2,928 versus 3,305 and 4,211). The mean fill rate is 0.18 points lower than for SAP-derived (98.11% versus 98.30%; bootstrap interval -0.54 to $+0.10$) and 0.11 points lower than for the SLT-informed OR. The mean cost difference against SAP-derived is -377 (95% interval -651 to -154); the LLM is cheaper than SAP-derived in 241 of 346 pairs and, in 226 pairs, simultaneously no more

expensive and no worse in fill. The LLM incurs more stockout days in total (910 versus 804 and 831), and aggregate fill is 78.87% versus 79.34% and 79.15%. Aggregate fill is dominated by a few high-demand materials: in the unconstrained simulation of run 1, excluding the five largest materials raises all three to 94.6–95.8% (Figure 2).

The cost decomposition shows where the saving comes from. Relative to SAP-derived parameters the LLM carries about 17% less average inventory (16,256 versus 19,565 units) and has about 16% lower holding cost, while it places about the same number of orders per year (17.3 versus 17.6). The saving is therefore a trade of inventory for service: the LLM holds less stock and accepts about 13% more stockout days. It is not the consequence of a better ordering rhythm, and it is not free of service cost. Section 6.8 shows that, once shortages carry a cost, this trade no longer favours the LLM.

Table 7: Hard-limit replay, common-feasible cohort ($n = 346$), working-day receipts, zero shortage penalty. Cost is the mean material-horizon holding-plus-ordering cost; average inventory is in units; orders are per year.

Arm	Mean cost	Holding	Ordering	Avg. inventory	Orders	Mean fill (%)	Aggr. fill (%)	Stockout days
SAP-derived	3,305	2,172	1,132	19,565	17.6	98.30	79.34	804
SLT-informed OR	4,211	2,608	1,603	21,614	24.5	98.23	79.15	831
ERP-floor OR	4,248	2,633	1,616	21,937	24.7	98.23	79.15	831
History-only OR	2,764	1,725	1,039	15,972	16.3	98.14	78.72	874
LLM (run 1)	2,928	1,824	1,104	16,256	17.3	98.11	78.87	910
LLM (run 2)	2,931	1,850	1,081	16,599	17.0	97.89	78.70	968
LLM (run 3)	2,929	1,841	1,088	16,958	17.0	98.16	78.90	890
LLM without ERP	3,014	1,902	1,112	16,507	17.3	98.06	78.89	890

6.3. Information asymmetry: what the LLM adds beyond the ERP inputs

Table 8 compares the arms in the hard-limit replay on the 346 pairs of the cohort, for all three runs. Three results stand out.

First, the SLT-informed OR and its ERP-floor variant are the most expensive arms (4,211 and 4,248). The ERP safety lead time inflates their effective lead time, hence their buffers; against them the LLM is about 30–31% cheaper in every run. This comparison therefore mainly measures how heavily ERP inputs over-buffer, not the quality of the LLM.

Demand concentration and fill sensitivity

Largest-demand materials strongly influence full-cohort aggregate fill.

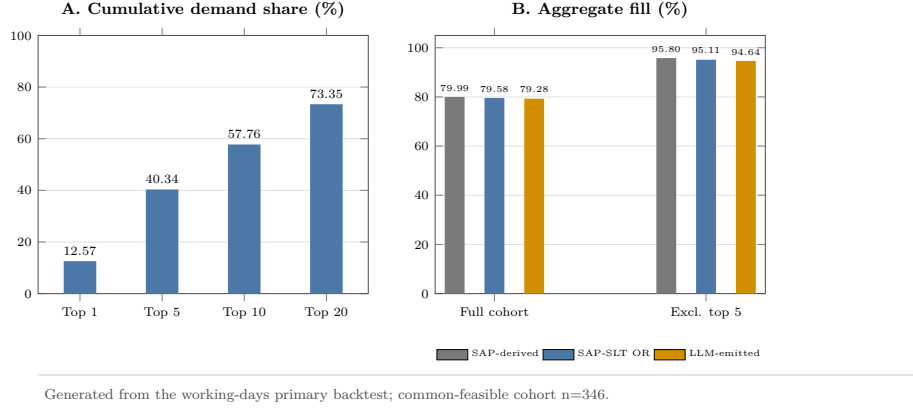


Figure 2: Demand concentration and aggregate-fill sensitivity (unconstrained simulation, run 1). Panel A: cumulative demand share of the largest materials. Panel B: aggregate fill for the full cohort and after excluding the five largest materials.

Second, the *history-only OR*, which uses no ERP inputs, has the lowest cost of all non-LLM arms (2,764 at a fill of 98.14%). Against it the LLM is 5.9–6.0% more expensive in all three runs (run 1: mean difference +164, 95% interval +91 to +240), with fill differences of -0.03 , -0.25 and $+0.01$ points whose intervals include zero in all three runs. The LLM is cheaper than the history-only OR in only 74–77 of 346 pairs. Part of the LLM’s lower cost relative to ERP therefore reflects the weakness of the ERP benchmarks and not LLM superiority.

Third, the LLM is nonetheless cheaper than ERP-derived parameters (11.3–11.4%) at a mean fill that is lower by 0.14–0.40 points (the interval excludes zero only in run 2), so it is useful when the alternative is the status quo, and it approaches a competent OR benchmark without code or training. Whether the LLM relies on the ERP planning parameters it is given is tested in the ablation of Section 6.4.

Table 8: Hard-limit replay: mean cost and mean fill by arm, $n = 346$; LLM cost relative to each arm (negative = LLM cheaper).

Arm	Mean cost	Mean fill (%)	LLM cost vs arm (%)		
			Run 1	Run 2	Run 3
SAP-derived	3,305	98.30	-11.4	-11.3	-11.4
SLT-informed OR	4,211	98.23	-30.5	-30.4	-30.5
ERP-floor OR	4,248	98.23	-31.1	-31.0	-31.1
History-only OR	2,764	98.14	+5.9	+6.0	+5.9
LLM (runs 1, 2, 3)	2,928 / 2,931 / 2,929	98.11 / 97.89 / 98.16			
LLM without ERP (ablation)	3,014	98.06	-8.8 [†]		

[†] Relative to SAP-derived; other comparisons are in Section 6.4.

6.4. LLM without ERP planning parameters (ablation)

In this ablation the LLM receives exactly the same material histories, cost parameters, MOQ, service target and storage limit as in runs 1–3, but the ERP safety stock and ERP safety lead time are blanked in the uploaded CSV and marked “not supplied” in the inline parameter block; the system prompt file is unchanged. It tests whether the LLM anchors on the ERP planning parameters it is given, and therefore whether the comparison with the ERP-informed arms is confounded by information. The ablation run has the same structure as runs 1–3: all 365 pairs returned a valid, storage-compliant policy.

Policy values. Withholding the ERP parameters changes the LLM’s policies only slightly beyond its ordinary run-to-run variation. The median ratios of safety stock and reorder point to run 1 are both 1.00 (346 common pairs). The reorder point differs by more than 10% in 54.6% of pairs, against 48.3% between runs 1 and 3 and 48.0% between runs 2 and 3 with the ERP parameters supplied; the safety stock differs by more than 10% in 11.0% of pairs, against 9.8% and 9.5%. The median absolute reorder-point difference is 128 units, against 90 and 84 between ERP-supplied runs. The mean ratio is slightly above one (1.04 for safety stock, 1.05 for the reorder point), so the LLM tends to buffer a little more when it has no ERP anchor.

427 *Outcomes.* In the hard-limit replay the LLM without ERP parameters
428 has a mean cost of 3,014 at a mean fill of 98.06%, against 2,928–2,931 and
429 97.89–98.16% in runs 1–3. It remains 8.8% cheaper than SAP-derived pa-
430 rameters (95% interval of the mean difference -566 to -61 ; fill -0.24 points,
431 interval including zero) and 28.4–29.1% cheaper than the SLT-informed and
432 ERP-floor ORs, but it is 9.0% more expensive than the history-only OR
433 (interval $+134$ to $+387$; fill -0.09 points; cheaper in 91 of 346 pairs). In the
434 unconstrained simulation the corresponding figures are -7.0% against SAP-
435 derived (interval -513 to $+16$, which includes zero), -23.2% and -23.8%
436 against the two ERP-informed ORs, and $+8.8\%$ against the history-only OR.
437 Withholding the ERP parameters thus raises the LLM’s cost by roughly 3
438 percentage points relative to the history-only OR and does not reverse any
439 ranking.

440 *By plant.* The pattern is the same as with ERP inputs: the LLM remains
441 cheaper than SAP-derived at Plants A (215 pairs, -10.8%) and B (65 pairs,
442 -21.9%) and more expensive at Plant C (66 pairs, $+41.6\%$ against $+36.2\%$
443 in run 1; cheaper in 8 pairs). The Plant C loss is therefore not caused by
444 the LLM leaning on ERP parameters either.

445 *Interpretation.* Two conclusions follow. First, the information-
446 asymmetry concern (that the LLM benefits from ERP inputs) does not
447 explain the LLM’s performance: with the ERP parameters removed the
448 LLM keeps most of its advantage over ERP-derived parameters, and the
449 gap to the history-only OR widens only moderately. Second, the ERP
450 parameters are of limited use to the LLM; most of its output is determined
451 by the demand and lead-time histories. A practical consequence is that the
452 LLM can be deployed for materials whose ERP planning data are missing or
453 stale. One limitation is that the ablation tests only the two ERP planning

454 fields; the LLM still sees unit cost, holding rate, ordering cost, MOQ, service
 455 target and storage limit, which all reference arms also use.

456 6.5. *Storage limits along simulated trajectories*

457 All 365 LLM policies satisfy the static rule $SS+Q \leq \text{max_storage_units}$.
 458 Along the simulated trajectories, however, the unconstrained simulation
 459 exceeds the limit on post-receipt on-hand stock for 183 pairs (52.9%) for
 460 the LLM (10.2% of simulated pair-days; median peak 1.02 times the limit;
 461 maximum 2.52), 198 pairs (57.2%; 15.9% of days) for SAP-derived and 230
 462 pairs (66.5%; 23.4% of days) for the SLT-informed OR; no pair starts above
 463 the limit. The three LLM runs give 183, 181 and 178 pairs. Exceedances
 464 arise because inventory on order and lead-time demand can add to stock
 465 above $SS+Q$; the static check is therefore necessary but not sufficient, and
 466 it applies to every method, not just the LLM.

467 Under the hard-limit replay no arm exceeds the limit by more than one
 468 unit (rounding). Enforcing the limit raises the LLM mean cost from 2,850
 469 to 2,928 and lowers its mean fill by 0.04 points (98.15% to 98.11%); SAP-
 470 derived rises from 3,184 to 3,305 and the SLT-informed OR from 3,852 to
 471 4,211, with the latter placing 24.5 instead of 15.8 orders on average. The
 472 ranking among arms is unchanged, and the LLM’s relative advantage over
 473 SAP-derived is 11.4% instead of 10.5%. We therefore use the replay as the
 474 primary evaluation.

475 6.6. *Plants and demand regimes*

476 The average hides a strong plant effect (Table 9, hard-limit replay, run 1).
 477 The LLM is 14.1% cheaper than SAP-derived in Plant A (cheaper in 179 of
 478 215 pairs) and 22.7% cheaper in Plant B (54 of 65), but 36.2% *more* expensive

in Plant C, where it is cheaper in only 8 of 66 pairs. The Plant C loss is similar in the other full runs (+32.2% and +31.5% in runs 2 and 3) and larger without the ERP parameters (+41.6%). Plant C is also the plant whose calendar we repaired; Section 6.8 shows that the loss persists under every working-day reconstruction in which Plant C can be scored. Against the history-only OR the LLM is 7.2% costlier in Plant A, 3.6% in Plant B and 6.7% in Plant C (run 1), so no plant favours the LLM over a plain history-based (r, Q) policy. Note that the SLT-informed and history-only ORs give identical results in Plant C, because the ERP safety lead time is zero for all 66 Plant C pairs.

By demand regime, the 33 lumpy materials show the largest cost reduction against SAP-derived (−25.4% in run 1; −23.7% to −25.2% in the other runs), whereas the 313 smooth materials show −9.8% at a 0.24-point lower mean fill (98.64% versus 98.88%). The service effect for lumpy materials is not consistent (mean fill +0.38 points in run 1, −0.29 in run 2), and the history-only OR is again 9% cheaper than the LLM there while achieving a much lower aggregate fill (54.3% versus 57.0% for the LLM and 65.8% for SAP-derived), so part of any cost saving on lumpy materials is bought with service. With only 33 lumpy materials we regard this as indicative only. By demand variability (terciles of the coefficient of variation), the LLM is 15.7% cheaper than SAP-derived at equal fill in the lowest tercile, 7.5% cheaper with a 0.68-point lower mean fill in the middle tercile (98.10% versus 98.78%), and 12.9% cheaper in the highest tercile.

Table 9: Mean simulated cost per pair and mean fill by plant, run 1, hard-limit replay (common-feasible cohort). The SLT-informed and history-only ORs coincide in Plant C.

Plant (<i>n</i>)	LLM cost	SAP-derived cost	SLT-informed OR cost	History-only OR cost	LLM vs SAP	Cheaper pairs	Fill LLM / SAP (%)
All (346)	2,928	3,305	4,211	2,764	−11.4%	241	98.11 / 98.30
A (215)	2,391	2,782	3,779	2,231	−14.1%	179	99.18 / 99.37
B (65)	4,916	6,360	7,329	4,746	−22.7%	54	96.23 / 96.47
C (66)	2,719	1,997	2,549	2,549	+36.2%	8	96.50 / 96.61

502 6.7. Run-to-run stability

503 Across the three runs the aggregate results are very similar: LLM cost
 504 relative to SAP-derived is -10.5% , -9.8% and -10.2% in the unconstrained
 505 simulation, and -11.4% , -11.3% and -11.4% in the hard-limit replay, so
 506 the spread is below one percentage point. The distributions of policy values
 507 are near-identical (median reorder point 1,584, 1,582 and 1,466; median or-
 508 der quantity 2,737, 2,777 and 2,693; median safety stock 683, 688 and 669;
 509 Figure 3). Individual policies, however, vary. Comparing runs 2 and 3 on
 510 the 365 pairs, all three policy values are identical for only 36 pairs; every
 511 policy resolves to the (r, Q) family, and the median absolute differences are
 512 81 units for the reorder point, 6 for the order quantity and 3 for the safety
 513 stock. A few pairs vary strongly (for one material the reorder point was
 514 14,882, 14,947 and 29,218 units in the three runs; for two others the safety
 515 stock was zero in one run and 660 or 1,075 units in the others). The practical
 516 implication is that one generation per material is a noisy estimate whose ag-
 517 gregate behaviour is stable but whose individual values should be reviewed.
 518 Non-determinism of this kind is a documented property of LLMs [16], so
 519 that repeated-run reporting should be standard.

520 6.8. Sensitivity analyses

521 *Plant C calendar..* Plant C’s source working-day flag is all zero, which re-
 522 quired reconstruction. We re-scored all policies (hard-limit replay) under
 523 four alternatives: any-plant, strict-all, exclusion of Plant C and calendar-
 524 day arrivals. In every variant in which Plant C can be scored (any-plant and
 525 calendar-day arrivals), the LLM remains 31–35% more expensive than SAP-
 526 derived in Plant C across the three full runs. The cohort-wide advantage
 527 over SAP-derived is -10.8% to -11.1% in these variants (primary calen-

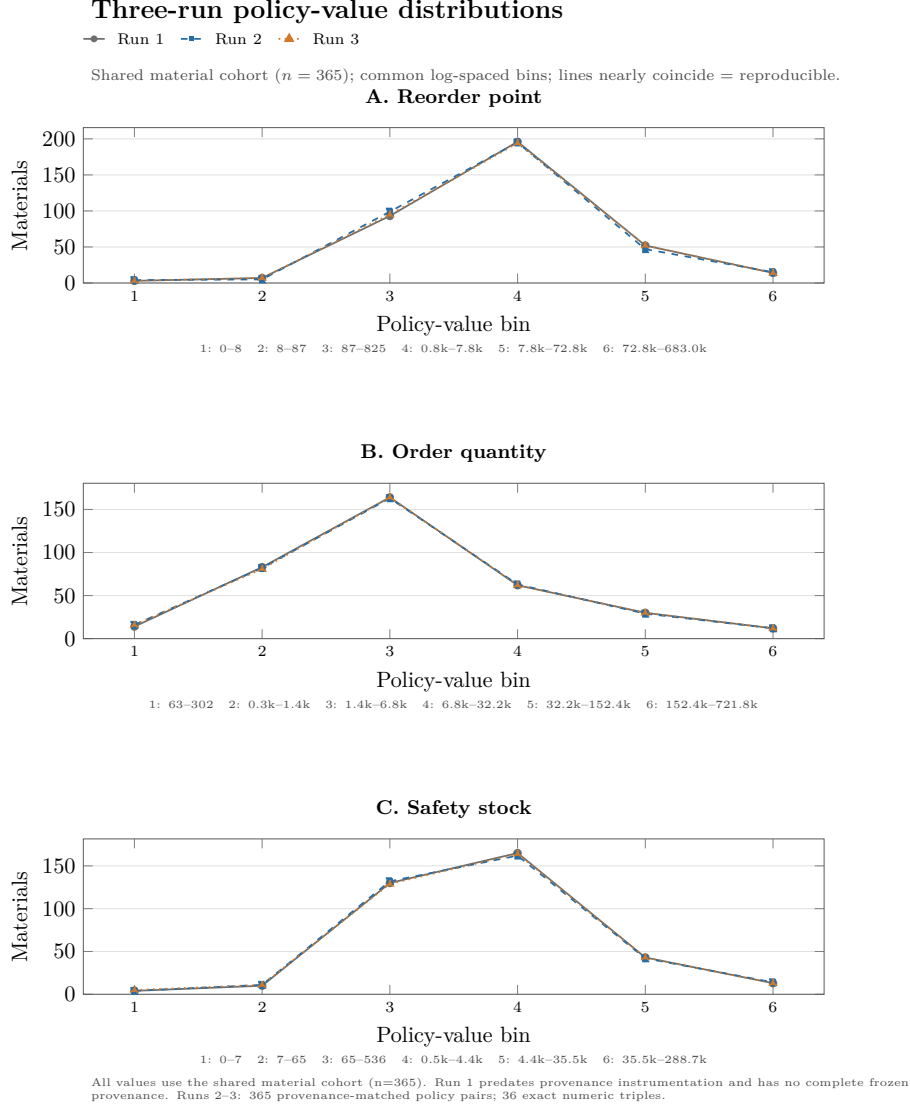


Figure 3: Policy-value distributions across the 365 shared pairs for runs 1, 2 and 3 (materials per common log-spaced bin); near-coincident lines indicate that aggregate behaviour is reproducible.

528 dar: -11.3% to -11.4%), and the cost excess over the history-only OR is
529 5.6–6.1%. The strict and exclusion variants (280 pairs) drop Plant C and
530 show a larger LLM advantage over SAP-derived (-17.0% to -17.6%) and

a 5.8–6.6% excess over the history-only OR. The Plant C result is therefore not an artefact of our calendar repair. One residual risk remains: the LLM saw demand series for Plant C that were themselves built on the repaired calendar, and only a re-generation with alternative inputs could remove that dependence.

Shortage penalty. With a positive lost-sales penalty the zero-penalty cost ranking changes. Table 10 gives mean total cost per pair (holding, ordering and shortage cost) in the hard-limit replay for penalties of zero, one and five times the unit cost. The SAP-derived parameters become the cheapest arm as soon as shortages are priced, because they carry more inventory and fewer stockouts. Run 1 of the LLM then costs 2.4% more than SAP-derived at both one and five times the unit cost; it remains cheaper than the SLT-informed OR (1.2% lower at both multiples) and the ERP-floor OR, and slightly cheaper than the history-only OR (0.8% lower at both multiples). The LLM’s cost advantage over ERP-derived parameters is therefore specific to the case in which service shortfalls are not charged, and the service deficit is a real cost when shortages are priced.

Table 10: Mean total cost per pair in the hard-limit replay (346 pairs) when lost sales are charged at a multiple of the unit cost.

Penalty	SAP-derived	SLT-informed OR	ERP-floor OR	History-only OR	LLM run 1	LLM run 2	LLM run 3	LLM no ERP
0×	3,305	4,211	4,248	2,764	2,928	2,931	2,929	3,014
1×	1,634,136	1,694,183	1,694,220	1,687,191	1,673,003	1,683,280	1,671,759	1,689,961
5×	8,157,462	8,454,072	8,454,109	8,424,899	8,353,306	8,404,676	8,347,082	8,437,748

Diagnostic outliers and calendar arrivals. In the unconstrained simulation of run 1, removing the 52 diagnostic outliers (314 pairs; coefficient of variation error above 0.75 or safety stock outside half to twice the reference) leaves LLM, SAP-derived and SLT-informed OR aggregate fill at 79.96%, 80.38% and 79.99%, with mean cost 2,864, 3,098 and 3,748, so the ordering

553 of the arms is unchanged. The findings thus do not depend on a few extreme
554 cases.

555 7. Managerial Insights

556 1. **Position the tool as a reviewed decision support system, not an**
557 **autonomous policy engine.** Single generations differ at the material
558 level, and roughly 6–8% of outputs carry a free-text escalation flag. The
559 gate-simulate-review structure (Section 4.3) captures most of the value
560 of the LLM (a candidate policy for every material, with a justification)
561 while leaving the decision to the planner.

562 2. **Benchmark against a history-only OR policy before claiming**
563 **value.** ERP-derived parameters and ERP-informed OR policies are weak
564 benchmarks: safety stocks are often stale or zero and safety lead times
565 inflate the effective lead time. A planning team that evaluates an LLM
566 aid only against ERP will overestimate its benefit. The history-only (r, Q)
567 is cheap to compute and was cheaper than the LLM in our data.

568 3. **Where the LLM is useful.** Plants A and B (cost reductions of 14–23%
569 against ERP), lumpy materials (about 25% cheaper than ERP-derived
570 parameters, although with mixed service effects and only 33 materials)
571 and the many materials with no analyst time are the sweet spot. Plant C,
572 where the LLM loses, indicates that conditions with repaired calendars,
573 small samples or atypical lead-time behaviour need escalation.

574 4. **Verify storage along the trajectory.** The static rule $SS + Q \leq \text{limit}$
575 cannot protect a warehouse: 53–66% of pairs breach the limit in simula-
576 tion under every method. Operations should enforce a cap in the order-

577 release logic (order-up-to or truncation), or treat the limit as a design
578 requirement in the policy.

579 **5. Price shortages explicitly.** Without a shortage penalty the cost rank-
580 ing rewards under-stocking. With a penalty of one unit cost the LLM’s
581 cost advantage over SAP disappears (it is about 2.4% more expensive).
582 Planners should set service targets and penalties before reading the cost
583 comparison.

584 **6. Freeze and log the configuration.** Treat the system prompt as a
585 controlled file, log every call (input hash, time, session, raw response)
586 and re-run before changing the prompt. Our reproducibility argument
587 rests on this log, not on the platform’s model identifier, which the client
588 cannot see.

589 **7. Cost and latency.** Each LLM generation took about 7–20 minutes,
590 whereas the OR baseline takes milliseconds. The LLM is attractive where
591 analyst time and not compute is scarce, but a deployment needs batching
592 and monitoring. Periodic regeneration as demand and lead times change
593 would also support adaptable planning under uncertainty [73].

594 **8. Discussion, Limitations and Research Perspectives**

595 *What the evidence shows..* In a three-plant industrial dataset, a prompt-
596 configured LLM generates feasible, reviewable (r, Q) policies for every mate-
597 rial. They are cheaper than ERP-derived parameters in the simulation, but
598 they do not beat a history-only OR policy. We therefore describe InventOR
599 as a way to make a competent analyst’s workflow available at scale, not as a
600 method that outperforms optimization. The information-controlled evalua-

tion is what allows this conclusion; without the ERP-independent arm, the first-cut result (-11.4% against SAP and -30.5% against the SLT-informed OR in the hard-limit replay) would have been presented as superiority.

Design principles for LLM-based decision support.. The evidence supports six principles that go beyond this application and that we offer as a basis for design theory [7]. **DP1, bound the generator:** give the LLM the role of candidate generator with a prompt that fixes the method, the constraints and the output schema, and no authority to act. **DP2, verify deterministically and along time:** check every output with code, and check feasibility along the trajectory, because point checks of constraints can pass while the constraints are violated over time (Section 6.5). **DP3, show counterfactual evidence:** give the reviewer a simulated comparison with credible references, so that review is based on outcomes and not on the persuasiveness of the justification. **DP4, control information in evaluation:** compare against baselines that use the same information and against a baseline that uses none, and withhold incumbent inputs from the LLM in an ablation, otherwise a gain may merely reflect a weak benchmark (Sections 6.3, 6.4). **DP5, freeze and log:** treat the prompt as a controlled file, log every call and repeat the runs, because individual outputs vary (Section 6.7). **DP6, escalate by rule:** define in advance when proposals are trusted, escalated or rejected, to limit complacency [13] (Section 4.3).

Limitations.. (i) The data come from one manufacturer and three plants; generalization to other settings is not shown. (ii) The platform does not expose the model identifier or inference settings, so the model is operator-declared. (iii) The simulation excludes observed receipts, corrections and scrap and starts with an empty pipeline; a prospective field trial with plan-

ners is needed to assess realized benefit, which we do not claim. (iv) The cost model has zero shortage penalty in the primary results and standard holding and ordering costs; with a positive penalty the cost ranking changes. (v) The (r, Q) policy is a single-item, single-echelon family; multi-item coupling and shared capacity are outside the scope. (vi) Run-to-run variation of individual policies is substantial. (vii) Plant C required calendar reconstruction. (viii) No planner outcomes were measured and the dialog component is a prototype record, not an evaluated interface: we make no claim about usability, adoption, planner workload or decision quality, which a DSS evaluation in the field would need to establish. (ix) The reference arms are normal-approximation (r, Q) policies; we did not include an intermittent-demand-specific baseline (for example Croston- or SBA-based forecasts [47, 48]) or a trained data-driven or reinforcement-learning policy, so the comparison with the state of the art is incomplete, particularly for lumpy materials.

Research perspectives..

1. *Ensembling and consistency.* Aggregating several generations (median or consensus policies) may reduce the individual-policy noise while retaining the LLM’s justification; the copilot findings for spare parts [34] suggest this is promising.
2. *Hybrid generators.* Combining the history-only OR with the LLM (the LLM reviewing and annotating the OR policy, or choosing among candidates by regime), as proposed for LLM–OR integration [74], could capture the OR’s cost level and the LLM’s flags.
3. *Comparison with trained methods.* A benchmark that compares prompt-based generators with trained data-driven or reinforcement-learning poli-

652 cies [46, 35, 36] on identical data and information would clarify the trade-
653 off between setup effort and policy quality.

654 4. *Intermittent-demand and richer policy families.* Allowing order-up-to or
655 (s, S) policies and regime-specific estimators may help the lumpy and
656 Plant C pairs [50, 51].

657 5. *Planner-in-the-loop field trials.* Prospective studies should measure plan-
658 ner time, override rates and realized service and inventory, which our
659 simulation cannot, and should examine adoption drivers and barriers with
660 mixed-methods designs [20, 19, 5].

661 6. *Standards for information-controlled evaluation.* Studies of LLM-based
662 decision support systems should report an ERP-independent baseline, an
663 ablation that withholds the incumbent’s inputs, trajectory-level feasibil-
664 ity, and run-to-run variation.

665 9. Conclusion

666 We defined InventOR as a human-reviewed decision support system for a
667 recurring production-planning decision, the setting of replenishment parame-
668 ters per plant-material pair, and evaluated it against six design requirements
669 on a three-plant industrial dataset. A prompt-configured LLM generated
670 feasible, justified (r, Q) policies for every material without training, and a
671 deterministic gate, a hard-limit simulation and explicit review rules made
672 these proposals checkable by a planner. The proposals cost about 11% less
673 than ERP-derived parameters at slightly lower service, but about 6% more
674 than a history-only OR policy; when lost sales are priced at one unit cost
675 or more, ERP-derived parameters become the cheapest arm. Storage limits

that hold in a static check are exceeded along simulated trajectories for every method, and the LLM loses to ERP at one plant. The value of the system therefore lies in its role as an auditable candidate generator inside a reviewed process, benchmarked against history-only OR, with trajectory-level storage checks, explicit shortage pricing and a frozen, logged configuration, and not in out-optimizing a competent OR baseline.

For the wider production-research community, the contribution is a set of design principles (Section 8) that apply to any planning decision in which parameters are set item by item and an LLM could be asked to propose them: bound the generator, verify deterministically and along time, show counterfactual evidence, control information in evaluation, freeze and log the configuration, and escalate by rule. These results are simulated-comparison evidence for an artifact; field evidence on planner behaviour and realized outcomes remains the central open step.

Appendix A. Formal Definitions

Reference calculation.. These are the standard textbook calculations [59, 68]. For pair i , let $\bar{d}_i$ and $\sigma_{D,i}$ be the mean and standard deviation of pre-cutoff working-day demand, L_i and $\sigma_{L,i}$ the mean and standard deviation of the lead time in working days, K_i the ordering cost, h_i the annual holding cost per unit (unit cost times holding rate), z_i the standard normal quantile of the service target, and 250 the working days per year. The order quantity is $Q_i = \max\{\text{MOQ}_i, \lceil \sqrt{2K_i \cdot 250\bar{d}_i/h_i} \rceil\}$. The lead-time-demand standard deviation is $\sigma_{LTD,i} = \sqrt{L_i\sigma_{D,i}^2 + \bar{d}_i^2\sigma_{L,i}^2}$, the safety stock $SS_i = \lceil z_i\sigma_{LTD,i} \rceil$ and the reorder point $R_i = \lceil \bar{d}_iL_i^{\text{eff}} + SS_i \rceil$, where L_i^{eff} is the mean lead time for the history-only OR and the mean lead time plus the ERP safety lead

time for the SLT-informed OR and the ERP-floor OR. The SAP-derived arm uses the ERP safety stock and the same Q_i . All policies are projected to $SS_i + Q_i \leq \text{max_storage_units}_i$ when $\text{MOQ}_i + SS_i$ fits.

Simulation.. On each post-cutoff working day, demand is served from on-hand stock, unmet demand is lost, and an order of size Q_i is placed when the inventory position (on-hand plus on order) is at or below R_i and arrives after the rounded mean lead time in subsequent working observations. Observed receipts, corrections and scrap are ignored. The total cost is $TC_i = h_i \bar{I}_i T / 250 + K_i N_i$, where $\bar{I}_i$ is the mean on-hand stock over the T simulated working days and N_i the number of orders. Fill rate is served divided by demanded units. In the hard-limit replay an order is truncated to $\text{max_storage_units}_i$ minus the inventory position, and no order is placed if the truncated quantity is zero.

Appendix B. Reproducibility Record

Prompt file, runner code, evaluation scripts, aggregate outputs and per-call input hashes are distributed with the code release (Data and Code Availability). The three runs and the ablation use the same prompt file; the ablation differs only in that the two ERP planning fields are blanked in the uploaded data and marked “not supplied” in the inline block. The deployed system prompt is the file `system_prompt.txt` in the release. The platform does not expose the model identifier or inference settings to the client; the model is therefore operator-declared (Anthropic Claude Sonnet 4.6 on an enterprise platform with a code-interpreter tool) and not independently verified. For each call the release records the input data, a hash of the input, the UTC time, the session identifier and the raw response.

726 **Author Contributions**

727 Conceptualization, Methodology, Software, Formal analysis, Validation,
728 Visualization, Writing – original draft, Writing – review and editing: Aris
729 Dressino.

730 **Funding**

731 This research did not receive any specific grant from funding agencies in
732 the public, commercial, or not-for-profit sectors.

733 **Declaration of Competing Interest**

734 The author declares no known competing financial interests or personal
735 relationships that could have appeared to influence the work reported in this
736 paper.

737 **Data and Code Availability**

738 The raw operational data and material-level artifacts cannot be publicly
739 released because they contain commercially sensitive material, supplier
740 and planning information. A confidentiality-screened release package
741 containing analysis code, the anonymized variable schema, aggregate
742 evaluation results, manuscript sources and prompt files is available at
743 github.com/DressPD/inventor_public. A versioned Zenodo archive of
744 the release is available at <https://doi.org/10.5281/zenodo.22262052>. A
745 preprint of an earlier version is archived at <https://doi.org/10.5281/zenodo.22262180>.
746 Supplementary File S1 contains the system prompt used for all runs.

747 **Acknowledgments**

748 The author thanks the industrial collaborators and material-control ex-
749 perts who contributed domain knowledge and access to anonymized opera-
750 tional data, and the enterprise data and IT teams for supporting data access
751 and prototype execution. No formal planner evaluation is claimed. Any
752 errors or omissions remain the responsibility of the author.

753 **Declaration of Generative AI and AI-Assisted Technologies in the** 754 **Writing Process**

755 During manuscript preparation and revision, the author used OpenAI
756 GPT-5.6 and Anthropic Claude models through the OpenCode coding assis-
757 tant. They supported language editing, restructuring and redrafting of the
758 text, consistency checks, citation verification against Crossref metadata, La-
759 TeX proofing, and the writing and execution of the analysis scripts that pro-
760 duce the evidence tables (for example the capacity-trajectory, information-
761 asymmetry and sensitivity analyses). These writing and coding assistants
762 are distinct from the Anthropic Claude Sonnet 4.6 deployment that is the
763 object of evaluation in the research workflow. The author specified the anal-
764 yses, reviewed and edited all content and results after use, and takes full
765 responsibility for the content of the published article. Figures were rendered
766 deterministically from author-reviewed LaTeX and Python source; genera-
767 tive AI did not directly create or manipulate image content.

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